HT Week 8 Model

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March 2026

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1 Introduction

This document outlines a game-theoretical model for pharmaceutical innovation and pricing to explain the empirical puzzle observed in the patent market following the expansion of Medicaid under the Affordable Care Act. The model captures the stochastic nature of R&D, the specific procurement mechanism (PDL bidding) used by state agencies, and how current procurement outcomes dictate future innovation incentives.

Figure 1 summarises the sequence of play, mapping the innovation decision in Period 1 to the pricing outcomes, which subsequently define the firm's state for future investment decisions in Period 2.

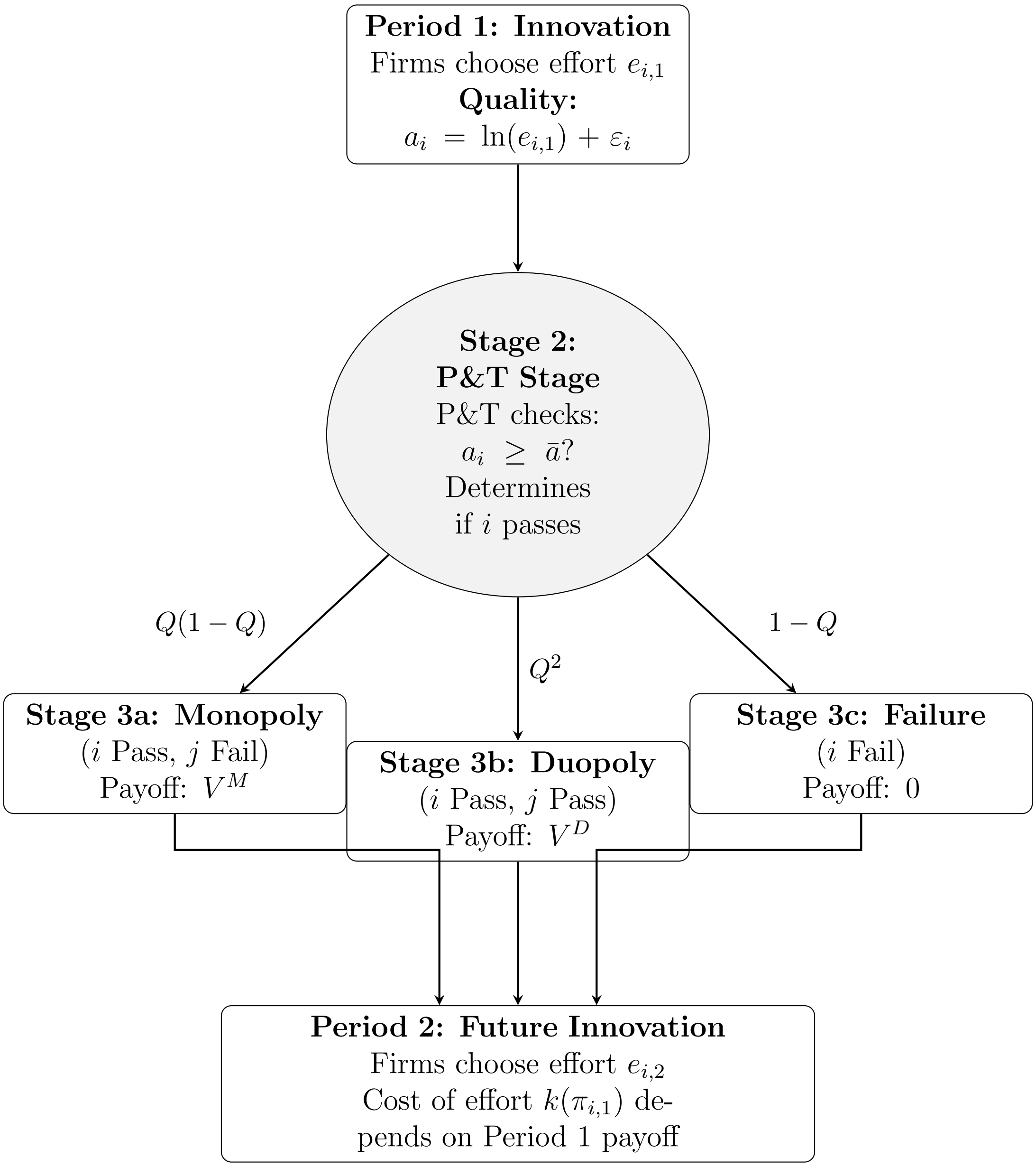

Figure 1: The Sequence of Play: Innovation, P&T Approval, Pricing, and Future Investment

2 Stage 1: The Innovation Decision (Effort)

First, firms make investment decisions for a drug they wish to market. Innovation is modelled to require effort - this is costly but improves the probability of clinical success.

2.1 The Production Technology

The realised clinical quality a_i is a function of the effort exerted by the firm and a stochastic component ε_i:

$$a_i = \ln(e_i) + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2) \tag{1}$$

where ε_i represents the inherent uncertainty of the R&D process.

2.2 The Cost of Effort

Firms i and j simultaneously choose R&D effort levels $e_i \geq 0$. Effort is assumed to be costly and convex:

$$C(e_i) = \frac{1}{2} k e_i^2 \tag{2}$$

3 Stage 2: The P&T Stage

Once effort is exerted it is now a sunk cost, nature determines the clinical outcome of each firm's drug a_i, i.e. whether the drug passes the P&T stage.

3.1 The Approval Condition

A drug is clinically approved (passes the P&T stage) if its quality a_i exceeds a regulatory threshold $\bar{a}$:

$$\text{P\&T Stage}_i = \begin{cases} \text{Approved} & \text{if } a_i \geq \bar{a} \\ \text{Not Approved} & \text{if } a_i < \bar{a} \end{cases} \tag{3}$$

3.2 The Probability of Success (Q)

The probability of passing, denoted $Q(e_i)$, is endogenous to effort:

$$\begin{aligned} Q(e_i) &= \Pr(\ln e_i + \varepsilon_i \geq \bar{a}) && (4) \\ &= \Pr(\varepsilon_i \geq \bar{a} - \ln e_i) && (5) \\ &= 1 - \Pr(\varepsilon_i < \bar{a} - \ln e_i) && (6) \\ &= 1 - \Phi\left(\frac{\bar{a} - \ln e_i}{\sigma}\right) && (7) \end{aligned}$$

where $\Phi(\cdot)$ is the standard normal CDF.

This defines the transition probabilities into Stage 3.

4 Stage 3: The Pricing Subgame

Firms that pass Stage 2 compete for market access. A rebate wedge $\tau(M)$ is introduced, where $\tau'(M) > 0$ - this represents the state's increasing monopsony power as their market size M grows.

The game resolves into one of three mutually exclusive states.

4.1 Stage 3a: The Monopoly State

Condition: Firm i passes while firm j fails:

$$a_i \geq \bar{a} > a_j$$

Mechanism: Firm i is the sole player in the market and negotiates directly with the state.

Setup The firm faces a market demand M which is perfectly inelastic with respect to price. Because Medicaid enrollees are fully insured with negligible out-of-pocket cost-sharing, we assume demand is completely inelastic up to the population limit M, a standard assumption in the literature modelling public insurance procurement. However, the state imposes a maximum allowable price $\bar{p}$ and the firm also pays some rebate $\tau(M)$.

Optimisation Problem The firm chooses price p to maximise profit:

$$\max_{p} \pi_i = M(p - c_i - \tau(M)) \quad \text{s.t.} \quad p \leq \bar{p} \tag{8}$$

Since demand is inelastic, the profit function is strictly increasing in p.

Equilibrium Result The firm sets the price at the binding cap: $p^* = \bar{p}$, and the resulting profit is:

$$\pi_i^{Mono}(c_i) = M(\bar{p} - c_i - \tau(M)) \tag{9}$$

Denote the expected value of this state across marginal costs as:

$$V^M = \mathbb{E}_c[\pi_i^{Mono}] \tag{10}$$

4.2 Stage 3b: The Duopoly State (Auction)

Condition: Both firms pass the P&T stage:

$$a_i \geq \bar{a}, \qquad a_j \geq \bar{a}$$

Mechanism: Firms compete in a first-price sealed-bid procurement auction.

Setup Each firm independently draws a marginal cost c_i, c_j from a common distribution $F(c)$ with support $[\underline{c}, \bar{c}]$. Each firm submits a sealed bid (net price) b_i. The lowest bidder wins PDL status and obtains the full market volume M minus the baseline statutory wedge $\tau(M)$.

$$\pi_i = \begin{cases} M\big(b_i - c_i - \tau(M)\big) & \text{if } b_i \leq b_j, \\ 0 & \text{if } b_i > b_j. \end{cases}$$

Equilibrium Derivation Assume that there exists a strictly increasing and differentiable symmetric Bayesian-Nash Equilibrium bidding strategy $b(c)$.

Step 1: The Probability of Winning Suppose firm i has cost c but considers submitting a bid $\tilde{b}$ corresponding to the optimal bid of type t (i.e., $\tilde{b} = b(t)$). Firm i wins if its bid is lower than the rival's bid $b(c_j)$:

$$b(t) < b(c_j) \iff t < c_j \tag{11}$$

The probability of winning is therefore:

$$\Pr(\text{win}) = \Pr(c_j > t) = 1 - F(t) \tag{12}$$

Step 2: The Optimisation Problem The firm chooses the target type t to maximise expected profit. Let $\tilde{c} = c + \tau(M)$ denote the effective cost including the statutory wedge.

$$\max_t \mathbb{E}[\pi(t, c)] = M(b(t) - c - \tau(M))[1 - F(t)] \tag{13}$$

Let $U(c)$ be the value function (maximum expected profit) for a firm with true cost c:

$$U(c) = \max_t M(b(t) - c - \tau(M))[1 - F(t)] \tag{14}$$

Step 3: Applying the Envelope Theorem Differentiation of the value function $U(c)$ with respect to cost c yields the partial derivative of the objective function evaluated at the optimal choice $t = c$ (truth-telling):

$$U'(c) = \left.\frac{\partial \mathbb{E}[\pi(t, c)]}{\partial c}\right|_{t=c} = -M[1 - F(c)] \tag{15}$$

Step 4: Solving for Expected Profit Integrating $U'(c)$ from c to $\bar{c}$, and noting that profit at the highest cost type $U(\bar{c}) = 0$:

$$U(c) = M \int_c^{\bar{c}} (1 - F(u))\, du \tag{16}$$

Step 5: Recovering the Bidding Function Equating the integral expression (16) with the definition of profit at the optimum yields the symmetric equilibrium bidding strategy:

$$b(c) = c + \tau(M) + \frac{\int_c^{\bar{c}} (1 - F(u))\, du}{1 - F(c)} \tag{17}$$

Equilibrium Result (Expected Payoff) Substituting the optimal bid, the ex-ante expected profit for entering Stage 3b is:

$$\Pi^{Duo}(c) = M \int_{c}^{\bar{c}} (1 - F(u))\, du \tag{18}$$

Denote the expected value of this state across marginal costs as:

$$V^D = \mathbb{E}_c[\Pi^{Duo}] \tag{19}$$

4.3 Stage 3c: The Failure State

Condition: Firm i fails:

$$a_i < \bar{a}$$

The firm fails to meet the clinical threshold required for PDL consideration. It cannot submit a bid.

Equilibrium Result The firm does not enter the market and earns zero revenue.

$$\pi_i^{Fail} = 0 \tag{20}$$

5 Equilibrium Analysis

Firstly, solve for the multi-period equilibrium by examining the optimal contemporaneous effort, and subsequently modelling how current procurement outcomes shape future innovation.

5.1 The Maximisation Problem

The firm chooses effort e_i to maximise total expected profit, where Ψ_i is the probability-weighted sum of payoffs in each state minus the cost of effort:

$$\max_{e_i} \Psi_i = \underbrace{Q(e_i)[1 - Q(e_j)]V^M}_{\text{Stage 3a Incentive}} + \underbrace{Q(e_i)Q(e_j)V^D}_{\text{Stage 3b Incentive}} - \frac{1}{2}ke_i^2 \tag{21}$$

where V^M and V^D are the expected values derived in the monopoly and duopoly cases, respectively.

5.2 Optimal Effort (e^*)

The First Order Condition (FOC) with respect to effort e_i is derived by differentiating Ψ_i:

$$\frac{\partial \Psi_i}{\partial e_i} = Q'(e_i)(1 - Q(e_j))V^M + Q'(e_i)Q(e_j)V^D - ke_i = 0 \tag{22}$$

This yields the following expression, where the marginal benefit of effort is equal to the associated marginal cost:

$$Q'(e_i) \left[(1 - Q(e_j))V^M + Q(e_j)V^D\right] = ke_i \tag{23}$$

The term in square brackets is the **expected continuation payoff given P&T approval**. In a symmetric equilibrium where $e_i = e_j = e^*$, this implicitly defines the optimal effort level e^*.

For this to be a strict global maximum, the Second Order Condition (SOC) must be strictly negative. Differentiating the FOC again with respect to e_i yields:

$$\frac{\partial^2 \Psi_i}{\partial e_i^2} = Q''(e_i) \left[(1 - Q(e_j))V^M + Q(e_j)V^D\right] - k < 0 \tag{24}$$

This condition holds under the assumptions of sufficient concavity in $Q(e)$ and k being sufficiently large.

5.3 Strategic Substitutes

The strategic relationship between the firms' efforts can be determined by analysing the slope of the best response function. Let $e_i^* = BR_i(e_j)$ be the implicitly defined best response function that solves the FOC:

$$\frac{\partial \Psi_i(BR_i(e_j), e_j)}{\partial e_i} = 0 \tag{25}$$

Totally differentiating this identity with respect to e_j yields:

$$\frac{d}{de_j}\left[\frac{\partial \Psi_i(BR_i(e_j), e_j)}{\partial e_i}\right] = 0 \tag{26}$$

$$\iff \quad \frac{\partial^2 \Psi_i}{\partial e_i^2}\frac{dBR_i(e_j)}{de_j} + \frac{\partial^2 \Psi_i}{\partial e_i \partial e_j} = 0 \tag{27}$$

$$\iff \quad \frac{dBR_i(e_j)}{de_j} = -\frac{\frac{\partial^2 \Psi_i}{\partial e_i \partial e_j}}{\frac{\partial^2 \Psi_i}{\partial e_i^2}} \tag{28}$$

The denominator is simply the SOC, which we know is strictly negative ($\frac{\partial^2 \Psi_i}{\partial e_i^2} < 0$). Therefore, the sign of the best response slope depends entirely on the sign of the cross-partial derivative:

$$\frac{\partial^2 \Psi_i}{\partial e_i \partial e_j} = Q'(e_i)\left[-Q'(e_j)V^M + Q'(e_j)V^D\right] = Q'(e_i)Q'(e_j)[V^D - V^M] \tag{29}$$

Since the monopoly value strictly exceeds the duopoly value ($V^M > V^D$), and marginal probabilities are positive ($Q' > 0$), this cross-partial derivative is strictly negative.

Because both the SOC and the cross-partial derivative are negative, it must follow that the slope of the best response function is negative ($\frac{dBR_i}{de_j} < 0$). Consequently, the R&D efforts of the two firms are **strategic substitutes**.

Intuitively, if firm j raises its effort, it becomes more likely to pass the clinical trials and the P&T stage. This increases the probability that both firms will enter the duopoly auction stage, where minimal profits (V^D) are realised. Since the expected profits shrink under such a situation, firm i's marginal benefit of effort decreases, prompting it to best-respond by lowering its own effort.

5.4 The "Incentive Trap" Mechanism

Now derive the effect of the expansion of Medicaid ($M \uparrow$) on innovation effort e^*. Innovation falls if the marginal benefit of effort decreases with market size.

The Mechanism

1. **Stage 3a (Monopoly):** From Equation (9), the value of the monopoly state is $V^M = M(\bar{p} - \bar{c} - \tau(M))$. The effect of expansion is unclear:

$$\frac{dV^M}{dM} = \underbrace{(\bar{p} - \bar{c} - \tau)}_{\text{Volume Effect (+)}} - \underbrace{M\tau'(M)}_{\text{Monopsony Effect (-)}} \tag{30}$$

2. **Stage 3b (Duopoly):** From Equation (18), V^D scales linearly with M. Expansion increases the value of the competitive state.

If the elasticity of the rebate $\epsilon_{\tau,M}$ is sufficiently large, the monopsony effect dominates, resulting in the value of Stage 3a (the monopoly) collapsing. Since the monopoly stage (3a) is the primary driver of R&D investment ($V^M \gg V^D$), the total incentive to innovate falls, explaining the negative coefficient observed in the data.

5.5 Institutional and Empirical Justification

A key assumption of this mechanism is that the state's monopsony wedge, $\tau(M)$, is strictly increasing in market size. While federal statutory rebates are fixed percentages, states actively negotiate 'supplemental rebates' with manufacturers in exchange for placement on the PDL. To maximise their negotiating leverage, a majority of U.S. states have joined Multi-State Purchasing Pools (such as the National Medicaid Pooling Initiative and the Sovereign States Drug Consortium). By aggregating the population under multiple states to increase their effective market size (M), these coalitions are able to tie the size of their negotiatedrebate to the volume of pooled lives.

This assumption is supported with the use of the State Drug Utilisation Data. By mapping individual states into their respective multi-state purchasing alliances to reflect the true negotiated market size, the per-prescription rebate value is regressed against the natural log of market size. When controlling for therapeutic class (ATC) and year fixed-effects, a highly significant positive relationship is found with $\beta = 5.51$ and $p < 0.001$. This confirms that larger Medicaid markets successfully exercise greater monopsony power to extract deeper rebates (τ), validating the theoretical condition that $\tau'(M) > 0$ and demonstrating how the ACA expansion directly compressed manufacturer margins.

5.6 State-Dependent Innovation

Empirically, firms that secure PDL placement tend to innovate more in subsequent periods. To capture this, the static procurement game can be embedded in a dynamic framework in which current procurement outcomes influence future innovation incentives. Following Ericson and Pakes (1995), firms' investment decisions depend on their continuation states, which are themselves determined by earlier competitive outcomes.

In this setting, Period 1 procurement outcomes map into different Period 2 states: failure, market access with competitive rents, and market access with higher rents. A natural microfoundation for this state dependence in pharmaceuticals is the internal-finance channel. Hal (2002) argues that R&D is particularly exposed to financing frictions because it is risky, intangible, and hard for outside investors to assess, making external finance expensive. In addition, successfully navigating the P&T process and rebate negotiations may build organisational and regulatory capability within the firm. Securing PDL rents therefore not only generates internal cash flow that relaxes financing constraints, but may also reduce the effective marginal cost of future R&D. Dynamic competition models (Budd et al., 1993) and step-by-step innovation models (Aghion et al., 2001) reinforce the broader point that a firm's current competitive position can shape its later investment incentives.

Formalising the Multi-Period Game To model this, allow the Period 2 cost of effort to depend on the realised payoff from Period 1. Let the Period 1 realised payoff for firm i be:

$$\pi_{i,1} \in \{V^M, V^D, 0\} \tag{31}$$

where $V^M > V^D > 0$.

Assuming that internal cash flow and accumulated regulatory capabilities lower the effective marginal cost of future R&D, the Period 2 effort cost function is:

$$C_2(e_{i,2}; \pi_{i,1}) = \frac{1}{2} k(\pi_{i,1}) e_{i,2}^2, \quad \text{where} \quad k'(\pi_{i,1}) < 0 \tag{32}$$

Because higher prior-period profits strictly reduce the cost parameter, we have:

$$k(V^M) < k(V^D) < k(0) \tag{33}$$

In Period 2, firm i solves an updated version of the First Order Condition (Equation 23), taking its new state-dependent cost parameter $k_i = k(\pi_{i,1})$ as given:

$$Q'(e_{i,2}) \left[(1 - Q(e_{j,2}))V^M + Q(e_{j,2})V^D \right] - k_i e_{i,2} = 0 \tag{34}$$

To determine how the optimal effort $e_{i,2}^*$ responds to a change in the cost parameter k_i, we apply the Implicit Function Theorem to the first-order condition. Let $\Psi_{i,2}$ denote the Period 2 objective function. Evaluating the partial derivatives yields:

$$\frac{\partial e_{i,2}^*}{\partial k_i} = -\frac{\frac{\partial^2 \Psi_{i,2}}{\partial e_{i,2} \partial k_i}}{\frac{\partial^2 \Psi_{i,2}}{\partial e_{i,2}^2}} = -\frac{-e_{i,2}}{\frac{\partial^2 \Psi_{i,2}}{\partial e_{i,2}^2}} \tag{35}$$

By the Second Order Condition, the denominator $\frac{\partial^2 \Psi_{i,2}}{\partial e_{i,2}^2}$ is strictly negative. Since effort $e_{i,2}$ is positive, the entire expression is strictly negative ($\frac{\partial e_{i,2}^*}{\partial k_i} < 0$).

Because optimal effort is strictly decreasing in the cost parameter k, and the cost parameter is strictly decreasing in Period 1 profits, the optimal effort in the Period 2 asymmetric equilibrium strictly follows the ranking of Period 1 states:

$$e_2^*(V^M) > e_2^*(V^D) > e_2^*(0) \tag{36}$$

Implications for the Incentive Trap This multi-period structure reinforces the primary mechanism of the model. If Medicaid expansion increases the rebate wedge $\tau(M)$ such that the monopoly prize V^M is compressed, the policy does not merely reduce the contemporaneous incentive to innovate. By suppressing the internal cash flows required to fund future R&D, the expansion structurally increases the future cost of effort $k(V^M)$, creating a compounded, multi-period decline in pharmaceutical innovation.

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